System Architecture & Processing Flowchart

Complete technical pipeline diagram for laser-cut metal stencils with multi-angle recovery, CLAHE contrast isolation, 6-key round-robin AI engine, and automatic catalog validation.

STEP 1: IMAGE CAPTURE & ARBITRARY ORIENTATIONS

Input Sources: Mobile App Camera, Handheld Terminal, or API Client.
Physical Realities: Arbitrary phone rotation (0°, 90°, 180°, 270°), mirrored reverse-side shots, laser stencil bridge gaps, harsh specular metal glare, and background floor colors showing through slits.

STEP 2: REST API GATEWAY (Flask Server on Render / Vercel)

Endpoint: POST /ocr | **Port:** 5000 / 443 HTTPS
Accepted Formats: Multipart Form (file key) OR JSON (image Base64).
Sub-millisecond Routing: Fast pre-check, payload decoding, and auto-dispatch to scanner.

STEP 3: LAB-SPACE CLAHE CONTRAST & DUAL-VIEW PIPELINE

CLAHE Contrast Enhancement: Converts BGR to LAB color space; applies Contrast Limited Adaptive Histogram Equalization (clip=3.0) on L-channel. Eliminates floor/background color confusion inside cutouts.
Dual-View Generation: Assembles **View 1 (Original)** and **View 2 (Horizontally Flipped)** into a single visual prompt.

STEP 4: MULTI-KEY ROUND-ROBIN & DYNAMIC FAILOVER AI ENGINE

Primary Vision Model: gemini-2.5-flash (ultra-fast 1.2s inference).
Fallback Chain: gemini-flash-latest ➔ gemini-3-flash-preview ➔ gemini-3.8-flash.
Active Key Pool: 6+ keys in circular round-robin. On HTTP 429 (rate limit) or 401, key gets 60s cooldown; request instantly auto-fails over to next active key.

STEP 5: IMPLEMENT CATALOG NORMALIZER & BRIDGE HEALING

Order Reversal: If code is read backwards (e.g. UP/.../GJ), tokens are automatically reversed.
State Normalization: Corrects bridge-cut 'G' (CJ, 6J, 0J ➔ GJ) or prepends missing state.
Model Catalog: Maps aliases (44G➔446, 367A➔3674, 44S➔445).
Serial Repair: Eliminates ghost digits from stencil bridge in '8', maps letters (O➔0, I➔1), ensures 5-digit format.

STEP 6: MINIMAL HIGH-SPEED JSON RESPONSE

Response Payload: {"status": "SUCCESS", "PARTno": "GJ/142/446/2026/11881/UP"}
Developer Experience: Clean, instant, and ready to consume by mobile app or ERP backend.

Physical Stencil Anatomy & Stencil Bridge Mechanics

Why standard OCR fails and how our specialized engine restores laser-cut implement codes:

Laser Through-Cut Slits Holes cut completely through sheet metal	Background floor, wooden pallets, rust, or shadows show through slits; model confuses colors.	**LAB CLAHE Preprocessing** isolates local luminance, removing background color hue and glare.
Stencil Font Bridge Tabs Metal tie-bars across 0, 4, 6, 8, 9, A, B, G	Broken character contours cause misreads: '4' → 'A', '6' → 'G', '8' → '38', 'G' → 'C'.	**Dual-View + Bridge Normalizer** heals broken contours and maps catalog aliases automatically.
Mirrored Reverse Shots Metal photographed from back side	Inverted text (e.g. 11541 → 12476) results in complete hallucination.	**Dual-View Generation** presents original + horizontally flipped views; AI auto-selects forward read.
Reversed Token Order Upside-down photo reads right-to-left	Produces 'UP/.../GJ'; regex expecting leading state fails to match.	**Auto-Reversal** detects trailing state ('GJ' at end) and flips token array into canonical order.

Live Production Endpoints & Integration Reference

Production Render URL	`https://s-ocr-3zbt.onrender.com/ocr`
Vercel Edge URL	`https://s-ocr.vercel.app/api/ocr`
HTTP Methods	**POST** (Process Image) | **GET** (Interactive Doc/Test)
Supported Payloads	Multipart form-data (file or image) | JSON (image base64)
Response Schema	`{"status": "SUCCESS", "PARTno": "GJ/142/446/2026/11881/UP"}`
Error Schema	`{"status": "FAILED", "PARTno": null, "error": "..."}`
Failover Pool	6+ Keys Round-Robin with 60s cooldown auto-switch on HTTP 429